11 Line 8 states that $\frac{a+b}{2} \geqslant \sqrt{ab}$ for $a,\ b \geqslant 0$. Explain why the result cannot be extended to apply in each of the following cases.

(i) One of the numbers a and b is positive and the other is negative. **[1]**

(ii) Both numbers a and b are negative. **[1]**